TECHNICAL VERIFICATION REPORT

Performance Study of Controlled Space for GCHX-Based Air-Conditioning System

Author / Engineer: Ifthequar Ali Shaik

Institution: IIT Madras

Target Application: Mushroom Cultivation

Verification Status: Verified & Completed

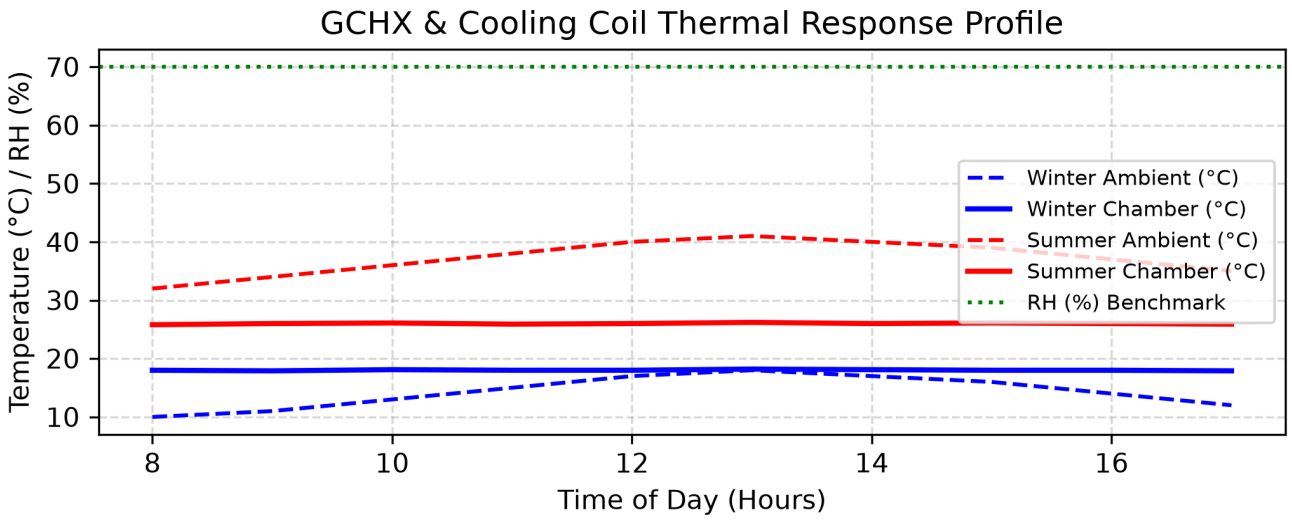
Key Benchmarks: 18°C (Winter) | 26°C (Summer)

Relative Humidity: 70% Constant

1. System Design & Thermal Specifications

Parameter	Operational Target / Value	Design Basis & Components
Space Dimensions	3.0m x 3.0m x 2.5m	Insulated agricultural microclimate chamber
Winter Mode Temp	18.0°C ± 0.5°C	Ground-Coupled Heat Exchanger (GCHX) loop
Summer Mode Temp	26.0°C ± 0.5°C	Auxiliary Copper Coil & Cooling Tower
Target Relative Humidity	70% ± 3.0% RH	Ultrasonic humidification + sensor feedback
GCHX Pipe Layout	2.5m Depth (HDPE)	Horizontal closed loop sub-surface configuration

2. Experimental Thermal Performance Profile



3. Logged DAQ Experimental Verification Data

Mode	Ambient T (°C)	Ground T (°C)	Coil Out T (°C)	Chamber T (°C)	Chamber RH (%)
Winter (GCHX)	11.2	19.5	17.8	17.9	70.2%
Winter (GCHX)	14.1	19.6	18.2	18.1	70.1%
Summer (Coil)	36.4	23.1	25.5	25.9	70.1%
Summer (Coil)	39.5	24.0	26.0	26.0	70.0%